

Question ID: a91a2b75

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

The function f is defined by $f(x) = -9x + 9$. What is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane?

Correct Answer: 9

Rationale

The correct answer is **9**. The y -intercept of the graph of $y = f(x)$ in the xy -plane is the point where the graph of $y = f(x)$ crosses the y -axis, which occurs at $x = 0$. It's given that the function f is defined by $f(x) = -9x + 9$. Substituting **0** for x and y for $f(x)$ in this equation yields $y = -9(0) + 9$, or $y = 9$. It follows that $y = 9$ when $x = 0$ and that the y -intercept of the graph of $y = f(x)$ in the xy -plane is **(0, 9)**. Therefore, the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane is **9**.